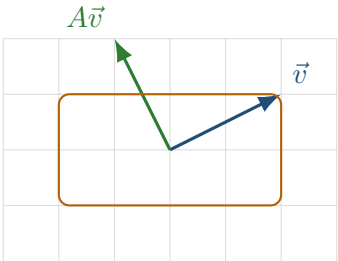
Matrices, Determinants, and Linear Systems

for Absolute Beginners

Lecture Notes for Computer Science Students



numbers $\longrightarrow$ tables $\longrightarrow$ transformations $\longrightarrow$ algorithms

These notes assume almost nothing. The goal is not to impress you with notation. The goal is to make matrices, determinants, and systems of linear equations feel predictable, mechanical, and useful.

Prepared as mathematical lecture notes

May 16, 2026

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1 How to Use These Notes

These notes are deliberately slow. Many students fail linear algebra not because the ideas are impossible, but because the notation begins too quickly. Here we will build everything from the smallest possible pieces.

Big idea

A matrix is just a rectangular table of numbers. The difficulty is not the table itself. The difficulty is learning what operations on the table mean.

Read these notes with paper and pencil. Whenever you see a computation, copy it and redo it. Linear algebra is learned by writing, not by watching.

1.1 What you should know before starting

You need only the following:

- how to add and multiply numbers,
- how to solve very simple equations such as $2x + 3 = 7$,
- how to substitute a number into a formula,
- how to keep track of signs.

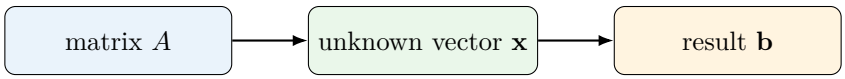
Everything else will be introduced gradually.

1.2 The central question

The central computational problem of these notes is:

$$A\mathbf{x} = \mathbf{b}.$$

At the beginning, this formula should look mysterious. By the end, it should feel like a compact way to write a system of ordinary equations.



2 From One Equation to Many Equations

2.1 One unknown

Start with a single equation:

$$2x = 10.$$

The solution is

$$x = 5.$$

This means: the number x that makes the equation true is 5.

Worked example

Solve

$$3x + 4 = 19.$$

First subtract 4 from both sides:

$$3x = 15.$$

Then divide by 3:

$$x = 5.$$

2.2 Two unknowns

Now consider

$$2x + y = 7.$$

This is not enough to determine one unique pair (x, y). For example:

$$(x, y) = (3, 1) \quad \text{works, because} \quad 2 \cdot 3 + 1 = 7.$$

But also

$$(x, y) = (2, 3) \quad \text{works, because} \quad 2 \cdot 2 + 3 = 7.$$

One equation with two unknowns usually has infinitely many solutions. To determine two unknowns, we usually need two independent equations.

2.3 A system of equations

A system is a group of equations that must all be true at the same time:

$$\begin{cases} 2x + y = 7, \\ x - y = 2. \end{cases}$$

A solution is a pair (x, y) that satisfies both equations.

Worked example

Check whether $(x, y) = (3, 1)$ solves

$$\begin{cases} 2x + y = 7, \\ x - y = 2. \end{cases}$$

First equation:

$$2 \cdot 3 + 1 = 7.$$

This is true. Second equation:

$$3 - 1 = 2.$$

This is also true. Therefore $(3, 1)$ is a solution.

Common mistake

A solution of a system must satisfy every equation. If it satisfies only one equation, it is not a solution of the system.

2.4 Why computer scientists care

Systems of linear equations appear in computer science in many places:

- graphics transformations,
- solving constraints,
- data fitting and regression,
- machine learning models,
- networks and graphs,
- simulations,
- coding theory and cryptography,
- numerical algorithms.

The same basic operations appear again and again: represent many numbers compactly, transform them, and solve for unknowns.

3 Vectors: Lists of Numbers

3.1 What is a vector?

For our purposes, a vector is an ordered list of numbers. A column vector looks like this:

$$\mathbf{v} = \begin{bmatrix} 3 \\ -1 \\ 4 \end{bmatrix}.$$

This vector has three entries. We say it is a vector in $\mathbb{R}^3$.

A two-dimensional vector is written as

$$\mathbf{u} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}.$$

This vector has two entries, so $\mathbf{u} \in \mathbb{R}^2$.

Definition

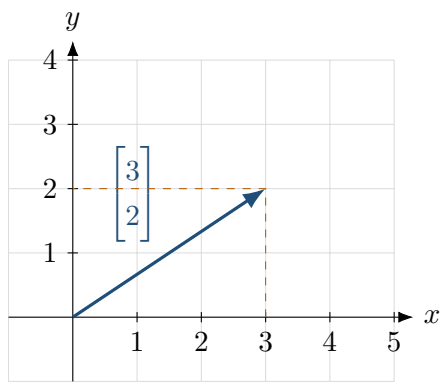
A vector in $\mathbb{R}^n$ is an ordered list of n real numbers:

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}.$$

The numbers $x_1, x_2, \ldots, x_n$ are called the entries or components of the vector.

3.2 Geometric picture in two dimensions

A vector in $\mathbb{R}^2$ can be drawn as an arrow from the origin.



The vector $\begin{bmatrix} 3 \\ 2 \end{bmatrix}$ means: move 3 units to the right and 2 units up.

3.3 Adding vectors

Vectors of the same size are added entry by entry:

$$\begin{bmatrix} 2 \\ 5 \end{bmatrix} + \begin{bmatrix} 1 \\ -3 \end{bmatrix} = \begin{bmatrix} 2+1 \\ 5+(-3) \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}.$$

Worked example

Compute

$$\begin{bmatrix} 4 \\ -1 \\ 7 \end{bmatrix} + \begin{bmatrix} -2 \\ 3 \\ 5 \end{bmatrix}.$$

Add corresponding entries:

$$\begin{bmatrix} 4+(-2) \\ -1+3 \\ 7+5 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 12 \end{bmatrix}.$$

3.4 Multiplying a vector by a number

A scalar is an ordinary number. Multiplying a vector by a scalar means multiplying every entry by that number:

$$3\begin{bmatrix} 2 \\ -5 \end{bmatrix} = \begin{bmatrix} 6 \\ -15 \end{bmatrix}.$$

Common mistake

Do not multiply only the first entry. The scalar multiplies every entry.

3.5 Linear combinations

A linear combination is an expression made from scalar multiples and sums of vectors.

For example:

$$2\begin{bmatrix} 1 \\ 0 \end{bmatrix} + 3\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}.$$

Big idea

Linear algebra is largely the study of linear combinations. Matrix multiplication, systems of equations, span, rank, and many algorithms are built from this idea.

4 Matrices: Rectangular Tables of Numbers

4.1 What is a matrix?

A matrix is a rectangular table of numbers:

$$A = \begin{bmatrix} 2 & 1 & 0 \\ -3 & 4 & 5 \end{bmatrix}.$$

This matrix has 2 rows and 3 columns. We say that its size is 2×3.

Definition

An $m \times n$ matrix has m rows and n columns:

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}.$$

The entry a_{ij} is in row i and column j.

column 2

$$\begin{bmatrix} 2 & 1 & 0 \\ -3 & 4 & 5 \\ 7 & 8 & 9 \end{bmatrix}$$

row 2

4.2 Reading entries

If

$$A = \begin{bmatrix} 2 & 1 & 0 \\ -3 & 4 & 5 \end{bmatrix},$$

then:

$$a_{11} = 2, \quad a_{12} = 1, \quad a_{13} = 0,$$
$$a_{21} = -3, \quad a_{22} = 4, \quad a_{23} = 5.$$

The first index is the row. The second index is the column.

Common mistake

The entry a_{23} means row 2, column 3. It does not mean row 3, column 2.

4.3 Why matrices are useful

Matrices let us store and manipulate structured information. Examples:

- a table of grades,
- pixel values in an image,
- connections in a graph,
- coefficients of a system of equations,
- coordinates of points,

- parameters of a model.

In computer science, a matrix is often both a mathematical object and a data structure.

4.4 Special matrices

A row matrix has one row:

$$\begin{bmatrix} 1 & 2 & 3 \end{bmatrix}.$$

A column matrix is a vector:

$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}.$$

A square matrix has the same number of rows and columns:

$$\begin{bmatrix} 1 & 4 \\ 2 & 5 \end{bmatrix}.$$

The zero matrix has all entries equal to zero:

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

The identity matrix has ones on the main diagonal and zeros elsewhere:

$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Big idea

The identity matrix plays the same role for matrices that the number 1 plays for ordinary multiplication.

5 Matrix Addition and Scalar Multiplication

5.1 Adding matrices

Matrices of the same size are added entry by entry:

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} + \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} = \begin{bmatrix} 1+5 & 2+6 \\ 3+7 & 4+8 \end{bmatrix} = \begin{bmatrix} 6 & 8 \\ 10 & 12 \end{bmatrix}.$$

Worked example

Compute

$$\begin{bmatrix} 2 & -1 & 0 \\ 4 & 3 & 5 \end{bmatrix} + \begin{bmatrix} 1 & 6 & 2 \\ -4 & 0 & 7 \end{bmatrix}.$$

Solution:

$$\begin{bmatrix} 3 & 5 & 2 \\ 0 & 3 & 12 \end{bmatrix}.$$

5.2 When addition is not allowed

You cannot add matrices of different sizes. For example,

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} + \begin{bmatrix} 5 & 6 & 7 \\ 8 & 9 & 10 \end{bmatrix}$$

is not defined, because the first matrix is 2×2 and the second is 2×3.

5.3 Scalar multiplication

Multiply every entry by the scalar:

$$-2 \begin{bmatrix} 1 & -3 \\ 0 & 4 \end{bmatrix} = \begin{bmatrix} -2 & 6 \\ 0 & -8 \end{bmatrix}.$$

Practice

Compute:

1. $\begin{bmatrix} 1 & 0 \\ -2 & 3 \end{bmatrix} + \begin{bmatrix} 4 & 5 \\ 6 & -1 \end{bmatrix}$
2. $3 \begin{bmatrix} 2 & -1 \\ 0 & 5 \end{bmatrix}$
3. $-\begin{bmatrix} 7 & 2 \\ -4 & 1 \end{bmatrix}$

6 Matrix-Vector Multiplication

6.1 The most important operation

Matrix-vector multiplication is the first operation that feels new. It is also one of the most important operations in all of applied mathematics and computer science.

Suppose

$$A = \begin{bmatrix} 2 & 1 \\ 3 & -1 \end{bmatrix}, \qquad \mathbf{x} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}.$$

Then

$$A\mathbf{x} = \begin{bmatrix} 2 & 1 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \end{bmatrix} = \begin{bmatrix} 2 \cdot 4 + 1 \cdot 5 \\ 3 \cdot 4 + (-1) \cdot 5 \end{bmatrix} = \begin{bmatrix} 13 \\ 7 \end{bmatrix}.$$

Each row of the matrix combines with the vector.

$$\begin{bmatrix} \boxed{\begin{matrix} 2 & 1 \end{matrix}} \\ \begin{matrix} 3 & -1 \end{matrix} \end{bmatrix} \begin{bmatrix} \boxed{\begin{matrix} 4 \\ 5 \end{matrix}} \end{bmatrix} = \begin{bmatrix} 2 \cdot 4 + 1 \cdot 5 \\ 3 \cdot 4 - 1 \cdot 5 \end{bmatrix}$$

6.2 Size rule

If A is an $m \times n$ matrix and $\mathbf{x}$ is a vector in $\mathbb{R}^n$, then $A\mathbf{x}$ is a vector in $\mathbb{R}^m$.

$$(m \times n)(n \times 1) = (m \times 1).$$

The inner sizes must match.

Common mistake

The product

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 7 \\ 8 \end{bmatrix}$$

is not defined because the matrix has 3 columns but the vector has only 2 entries.

6.3 Matrix-vector multiplication as a system of formulas

Let

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \qquad \mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix}.$$

Then

$$A\mathbf{x} = \begin{bmatrix} ax + by \\ cx + dy \end{bmatrix}.$$

So multiplying a matrix by a vector creates several linear expressions at once.

Big idea

A matrix is a machine. It takes an input vector and produces an output vector.

6.4 Column interpretation

Matrix-vector multiplication can also be understood as a linear combination of the columns of the matrix:

$$\begin{bmatrix} 2 & 1 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \end{bmatrix} = 4 \begin{bmatrix} 2 \\ 3 \end{bmatrix} + 5 \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 8 \\ 12 \end{bmatrix} + \begin{bmatrix} 5 \\ -5 \end{bmatrix} = \begin{bmatrix} 13 \\ 7 \end{bmatrix}.$$

This interpretation is extremely important for understanding systems.

7 Matrix-Matrix Multiplication

7.1 The basic rule

Matrix-matrix multiplication is repeated matrix-vector multiplication. If

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \qquad B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix},$$

then

$$AB = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}.$$

The entry in row i, column j of AB is obtained by multiplying row i of A with column j of B.

$$AB = \begin{bmatrix} 1 \cdot 5 + 2 \cdot 7 & 1 \cdot 6 + 2 \cdot 8 \\ 3 \cdot 5 + 4 \cdot 7 & 3 \cdot 6 + 4 \cdot 8 \end{bmatrix} = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}.$$

7.2 Size rule for matrix multiplication

If A is $m \times n$ and B is $n \times p$, then AB is $m \times p$:

$$(m \times n)(n \times p) = (m \times p).$$

Again, the inner sizes must match.

Worked example

Let A be 2×3 and B be 3×4. Then AB is defined and has size 2×4.
But BA is not defined here, because B is 3×4 and A is 2×3; the inner sizes 4 and 2 do

not match.

7.3 Matrix multiplication is not commutative

For ordinary numbers,

$$2 \cdot 3 = 3 \cdot 2.$$

For matrices, usually

$$AB \neq BA.$$

Sometimes one of the products may not even exist.

Common mistake

Never assume that $AB = BA$. This is one of the most common beginner errors.

7.4 Identity matrix

The identity matrix leaves vectors unchanged:

$$I_2 \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}.$$

It also leaves square matrices unchanged:

$$AI = A, \qquad IA = A.$$

8 Systems of Linear Equations as Matrices

8.1 The system

Consider the system

$$\begin{cases} 2x + y = 7, \\ x - y = 2. \end{cases}$$

The coefficient matrix is

$$A = \begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix}.$$

The unknown vector is

$$\mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix}.$$

The right-hand side vector is

$$\mathbf{b} = \begin{bmatrix} 7 \\ 2 \end{bmatrix}.$$

The system can be written as

$$A\mathbf{x} = \mathbf{b},$$

that is

$$\begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 7 \\ 2 \end{bmatrix}.$$

8.2 Why this notation is useful

Matrix notation is compact. Instead of writing many equations, we write one matrix equation.

But it is not just shorter. It lets us use general algorithms. The same method can solve two equations, three equations, or thousands of equations.

8.3 Augmented matrix

To solve a system, we often write the augmented matrix. For

$$\begin{cases} 2x + y = 7, \\ x - y = 2, \end{cases}$$

the augmented matrix is

$$\left[\begin{array}{cc|c} 2 & 1 & 7 \\ 1 & -1 & 2 \end{array}\right].$$

The vertical line separates coefficients from the right-hand side.

Big idea

The augmented matrix contains all numerical information from the system, but removes the variable symbols. This makes solving more mechanical.

9 Row Operations

9.1 The allowed moves

There are three elementary row operations:

1. Swap two rows.
2. Multiply a row by a nonzero number.
3. Add a multiple of one row to another row.

These operations are allowed because they do not change the solution set of the system.

9.2 Notation

We write R_1, R_2, R_3 for row 1, row 2, row 3.

Examples:

- $R_1 \leftrightarrow R_2$: swap rows 1 and 2.
- $R_2 \leftarrow 3R_2$: multiply row 2 by 3.
- $R_2 \leftarrow R_2 - 2R_1$: replace row 2 with row 2 minus twice row 1.

Common mistake

When you do $R_2 \leftarrow R_2 - 2R_1$, only row 2 changes. Row 1 stays the same.

9.3 Why row operations work

Suppose we have the equation

$$x + y = 5.$$

Multiplying both sides by 2 gives

$$2x + 2y = 10.$$

This equation has the same solutions. It is just the same information written differently.

Similarly, if two equations are true, then adding one equation to another gives another true equation. Row operations use this idea.

10 Gaussian Elimination: The Main Algorithm

10.1 Goal

Gaussian elimination changes a system into an easier system, usually triangular.

Example:

$$\begin{cases} 2x + y = 7, \\ x - y = 2 \end{cases}$$

is not as easy as

$$\begin{cases} x - y = 2, \\ 3y = 3. \end{cases}$$

The second system gives $y = 1$, then $x = 3$.

10.2 A complete step-by-step example

Solve

$$\begin{cases} x + y = 5, \\ 2x - y = 1. \end{cases}$$

Write the augmented matrix:

$$\left[\begin{array}{cc|c} 1 & 1 & 5 \\ 2 & -1 & 1 \end{array}\right].$$

Use row 1 to eliminate the 2 below it:

$$\left[\begin{array}{cc|c} 1 & 1 & 5 \\ 2 & -1 & 1 \end{array}\right] \xrightarrow{R_2 \leftarrow R_2 - 2R_1} \left[\begin{array}{cc|c} 1 & 1 & 5 \\ 0 & -3 & -9 \end{array}\right].$$

The second row says

$$-3y = -9.$$

So

$$y = 3.$$

Substitute into the first equation:

$$x + y = 5 \quad \Rightarrow \quad x + 3 = 5 \quad \Rightarrow \quad x = 2.$$

Therefore

$$(x, y) = (2, 3).$$

Checklist

Gaussian elimination usually follows this pattern:

1. Choose a pivot.
2. Use the pivot to create zeros below it.
3. Move to the next column and repeat.
4. Use back substitution.

10.3 Another example with three unknowns

Solve

$$\begin{cases} x + y + z = 6, \\ 2x + y - z = 3, \\ x - y + 2z = 7. \end{cases}$$

Augmented matrix:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & 1 & -1 & 3 \\ 1 & -1 & 2 & 7 \end{array}\right].$$

Eliminate entries below the first pivot:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & 1 & -1 & 3 \\ 1 & -1 & 2 & 7 \end{array}\right] \xrightarrow[R_3 \leftarrow R_3 - R_1]{R_2 \leftarrow R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -1 & -3 & -9 \\ 0 & -2 & 1 & 1 \end{array}\right].$$

Now use row 2 to eliminate the entry below the second pivot. Since row 3 has -2 and row 2 has -1, use

$$R_3 \leftarrow R_3 - 2R_2.$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -1 & -3 & -9 \\ 0 & -2 & 1 & 1 \end{array}\right] \xrightarrow{R_3 \leftarrow R_3 - 2R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -1 & -3 & -9 \\ 0 & 0 & 7 & 19 \end{array}\right].$$

The last row says

$$7z = 19,$$

so

$$z = \frac{19}{7}.$$

The second row says

$$-y - 3z = -9.$$

Substitute $z = \frac{19}{7}$:

$$-y - 3 \cdot \frac{19}{7} = -9.$$

$$-y - \frac{57}{7} = -\frac{63}{7}.$$

$$-y = -\frac{6}{7}.$$

$$y = \frac{6}{7}.$$

The first row says

$$x + y + z = 6.$$

So

$$x + \frac{6}{7} + \frac{19}{7} = 6.$$

$$x + \frac{25}{7} = \frac{42}{7}.$$

$$x = \frac{17}{7}.$$

Thus

$$(x, y, z) = \left(\frac{17}{7}, \frac{6}{7}, \frac{19}{7}\right).$$

Common mistake

Fractions are not a sign that something went wrong. Linear systems often have fractional solutions.

11 Pivots and Echelon Form

11.1 What is a pivot?

A pivot is the leading nonzero entry used to eliminate other entries in its column.

In the matrix

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 4 \\ 0 & 5 & 6 & 7 \\ 0 & 0 & 8 & 9 \end{array}\right],$$

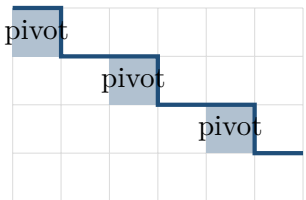
the pivots are 1, 5, and 8.

11.2 Echelon form

A matrix is in row echelon form if:

- all zero rows are at the bottom,
- each pivot is to the right of the pivot above it,
- all entries below each pivot are zero.

It looks like a staircase:



11.3 Back substitution

After reaching echelon form, solve from the bottom upward.

Example:

$$\begin{cases} x + 2y + z = 9, \\ 3y - z = 4, \\ 2z = 6. \end{cases}$$

Bottom equation:

$$2z = 6 \Rightarrow z = 3.$$

Middle equation:

$$3y - z = 4 \Rightarrow 3y - 3 = 4 \Rightarrow 3y = 7 \Rightarrow y = \frac{7}{3}.$$

Top equation:

$$x + 2y + z = 9.$$

$$x + 2 \cdot \frac{7}{3} + 3 = 9.$$

$$x + \frac{14}{3} + \frac{9}{3} = \frac{27}{3}.$$

$$x = \frac{4}{3}.$$

12 Three Possible Outcomes

A linear system can have exactly one solution, no solution, or infinitely many solutions.

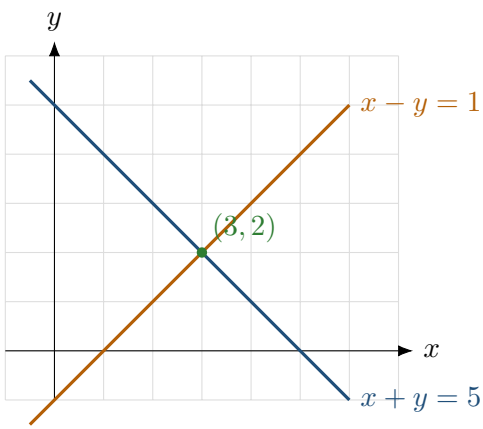
12.1 Exactly one solution

Example:

$$\begin{cases} x + y = 5, \\ x - y = 1. \end{cases}$$

This system has one solution: $(3, 2)$.

Geometrically, two lines intersect in one point.



12.2 No solution

Example:

$$\begin{cases} x + y = 5, \\ x + y = 2. \end{cases}$$

The same expression cannot equal both 5 and 2. During elimination, this may produce a row such as

$$\left[\begin{array}{cc|c} 0 & 0 & 3 \end{array}\right],$$

which means

$$0 = 3.$$

This is impossible.

Geometrically, the lines are parallel.

12.3 Infinitely many solutions

Example:

$$\begin{cases} x + y = 5, \\ 2x + 2y = 10. \end{cases}$$

The second equation is just twice the first. They contain the same information.

During elimination, this may produce a row such as

$$\left[\begin{array}{cc|c} 0 & 0 & 0 \end{array}\right],$$

which means

$$0 = 0.$$

This is always true and gives no new information.

Checklist

How to recognize the outcome:

- A row $[0\ 0\ \cdots\ 0 \mid c]$ with $c \neq 0$ means no solution.
- A pivot in every variable column usually means one solution.
- A missing pivot in a variable column and no contradiction means infinitely many solutions.

13 Reduced Row Echelon Form

13.1 Making the answer even clearer

Row echelon form is enough for solving, but reduced row echelon form makes the solution easier to read.

A matrix is in reduced row echelon form if:

- it is already in row echelon form,

- each pivot equals 1,
- each pivot is the only nonzero entry in its column.

Example:

$$\left[\begin{array}{cc|c} 1 & 0 & 3 \\ 0 & 1 & 2 \end{array}\right]$$

means directly

$$x = 3, \qquad y = 2.$$

13.2 Gauss-Jordan elimination

Gauss-Jordan elimination continues after Gaussian elimination by clearing entries above pivots as well.

Example:

$$\left[\begin{array}{cc|c} 1 & 2 & 7 \\ 0 & 1 & 3 \end{array}\right].$$

The second row says $y = 3$. To remove the 2 above the pivot, do

$$R_1 \leftarrow R_1 - 2R_2.$$

$$\left[\begin{array}{cc|c} 1 & 2 & 7 \\ 0 & 1 & 3 \end{array}\right] \xrightarrow{R_1 \leftarrow R_1 - 2R_2} \left[\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & 3 \end{array}\right].$$

So

$$x = 1, \qquad y = 3.$$

14 Determinants: First Meaning

14.1 Only for square matrices

The determinant is a number assigned to a square matrix.

For a 2×2 matrix

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix},$$

the determinant is

$$\det(A) = ad - bc.$$

Definition

For

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix},$$

$$\det(A) = ad - bc.$$

14.2 Example

Let

$$A = \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}.$$

Then

$$\det(A) = 2 \cdot 4 - 1 \cdot 3 = 8 - 3 = 5.$$

14.3 What does the determinant tell us?

The determinant tells us whether a square matrix is invertible and whether the corresponding system has a unique solution.

For a 2×2 system:

$$A\mathbf{x} = \mathbf{b},$$

if

$$\det(A) \neq 0,$$

then the system has exactly one solution.

If

$$\det(A) = 0,$$

then the system does not have exactly one solution. It may have no solution or infinitely many solutions.

Big idea

A nonzero determinant means the matrix has not collapsed space. A zero determinant means some direction has been collapsed.

15 Determinants as Area Scaling

15.1 The geometric idea

The columns of a 2×2 matrix are two vectors. These vectors form a parallelogram. The absolute value of the determinant is the area of that parallelogram.

Let

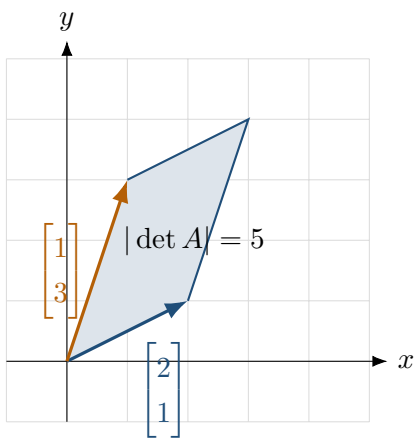
$$A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}.$$

The columns are

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix}, \qquad \begin{bmatrix} 1 \\ 3 \end{bmatrix}.$$

$$\det(A) = 2 \cdot 3 - 1 \cdot 1 = 5.$$

The area of the parallelogram is $|5| = 5$.



15.2 Zero determinant

If the determinant is zero, the parallelogram has area zero. This means the two column vectors lie on the same line.

Example:

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}.$$

$$\det(A) = 1 \cdot 4 - 2 \cdot 2 = 4 - 4 = 0.$$

The second column is twice the first:

$$\begin{bmatrix} 2 \\ 4 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

So the two vectors do not span a real two-dimensional area.

16 Determinants of Larger Matrices

16.1 The 3×3 case

For

$$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix},$$

the determinant can be computed by expansion along the first row:

$$\det(A) = a \det \begin{bmatrix} e & f \\ h & i \end{bmatrix} - b \det \begin{bmatrix} d & f \\ g & i \end{bmatrix} + c \det \begin{bmatrix} d & e \\ g & h \end{bmatrix}.$$

That is,

$$\det(A) = a(ei - fh) - b(di - fg) + c(dh - eg).$$

Worked example

Compute

$$\det \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{bmatrix}.$$

Expansion along the first row:

$$1 \det \begin{bmatrix} 1 & 4 \\ 6 & 0 \end{bmatrix} - 2 \det \begin{bmatrix} 0 & 4 \\ 5 & 0 \end{bmatrix} + 3 \det \begin{bmatrix} 0 & 1 \\ 5 & 6 \end{bmatrix}.$$

Now compute each 2×2 determinant:

$$1(1 \cdot 0 - 4 \cdot 6) - 2(0 \cdot 0 - 4 \cdot 5) + 3(0 \cdot 6 - 1 \cdot 5).$$

$$= -24 - 2(-20) + 3(-5) = -24 + 40 - 15 = 1.$$

So the determinant is 1.

16.2 Triangular matrices

A triangular matrix has zeros either below or above the main diagonal.

Example:

$$U = \begin{bmatrix} 2 & 5 & 1 \\ 0 & 3 & -2 \\ 0 & 0 & 4 \end{bmatrix}.$$

For triangular matrices, the determinant is the product of the diagonal entries:

$$\det(U) = 2 \cdot 3 \cdot 4 = 24.$$

This is one reason elimination is useful for determinants.

16.3 How row operations affect determinants

For square matrices:

- Swapping two rows changes the sign of the determinant.
- Multiplying a row by k multiplies the determinant by k.
- Adding a multiple of one row to another row does not change the determinant.

Common mistake

Row operations for solving systems and row operations for determinants are related, but the bookkeeping is different. If you multiply a row by k, the solution set is unchanged, but the determinant is multiplied by k.

17 Inverse Matrices

17.1 The idea of an inverse

For ordinary numbers, the inverse of 5 is $\frac{1}{5}$ because

$$5 \cdot \frac{1}{5} = 1.$$

For matrices, the inverse of a square matrix A is a matrix A^{-1} such that

$$AA^{-1} = I, \qquad A^{-1}A = I.$$

Not every matrix has an inverse.

17.2 Inverse of a 2×2 matrix

If

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix},$$

and

$$\det(A) = ad - bc \neq 0,$$

then

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

Worked example

Find the inverse of

$$A = \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}.$$

First compute the determinant:

$$\det(A) = 2 \cdot 4 - 1 \cdot 3 = 5.$$

Since $5 \neq 0$, the inverse exists:

$$A^{-1} = \frac{1}{5} \begin{bmatrix} 4 & -1 \\ -3 & 2 \end{bmatrix} = \begin{bmatrix} 4/5 & -1/5 \\ -3/5 & 2/5 \end{bmatrix}.$$

17.3 Solving with the inverse

If

$$A\mathbf{x} = \mathbf{b}$$

and A^{-1} exists, then

$$\mathbf{x} = A^{-1}\mathbf{b}.$$

However, in real computer programs we often do not compute the inverse explicitly. We usually solve the system directly using elimination or numerical methods.

Big idea

The inverse is conceptually important. Computationally, solving $A\mathbf{x} = \mathbf{b}$ is often better than calculating A^{-1}.

18 Rank: How Much Independent Information?

18.1 Intuitive meaning

The rank of a matrix is the number of independent directions or independent equations it contains.

After row reduction, the rank is the number of pivots.

Definition

The rank of a matrix is the number of pivot columns in its row echelon form.

18.2 Example

Reduce

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix}.$$

The second row is twice the first, so it does not add new information.

Perform elimination:

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix} \xrightarrow[R_3 \leftarrow R_3 - R_1]{R_2 \leftarrow R_2 - 2R_1} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & -1 & -2 \end{bmatrix}.$$

Swap rows 2 and 3:

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & -2 \\ 0 & 0 & 0 \end{bmatrix}.$$

There are two pivots, so

$$\operatorname{rank}(A) = 2.$$

18.3 Rank and systems

For a system $A\mathbf{x} = \mathbf{b}$:

- rank helps decide whether a solution exists,
- rank helps determine whether the solution is unique,
- rank tells us how many independent constraints we have.

19 Free Variables and Infinite Solutions

19.1 A simple example

Consider

$$\begin{cases} x + y + z = 4, \\ 2x + 2y + 2z = 8. \end{cases}$$

The second equation is twice the first. We really have only one independent equation with three unknowns.

The augmented matrix is

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 2 & 2 & 2 & 8 \end{array}\right].$$

Eliminate:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 2 & 2 & 2 & 8 \end{array}\right] \xrightarrow{R_2 \leftarrow R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

The equation is

$$x + y + z = 4.$$

Let

$$y = s, \qquad z = t.$$

Then

$$x = 4 - s - t.$$

So the solutions are

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 - s - t \\ s \\ t \end{bmatrix}, \qquad s, t \in \mathbb{R}.$$

19.2 What are free variables?

A free variable is a variable that can be chosen freely because its column has no pivot.

In the example above, y and z are free variables. Once we choose them, x is determined.

Common mistake

A free variable is not "missing". It means there are infinitely many solutions, described by parameters.

20 Cramer's Rule

20.1 A formula method

Cramer's rule solves small square systems using determinants. It is elegant, but it is usually not the best computational method for large systems.

For

$$\begin{cases} ax + by = e, \\ cx + dy = f, \end{cases}$$

define

$$D = \det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc.$$

If $D \neq 0$, then

$$x = \frac{\det \begin{bmatrix} e & b \\ f & d \end{bmatrix}}{D}, \qquad y = \frac{\det \begin{bmatrix} a & e \\ c & f \end{bmatrix}}{D}.$$

Worked example

Solve

$$\begin{cases} 2x + y = 7, \\ x - y = 2. \end{cases}$$

Here

$$D = \det \begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix} = 2(-1) - 1(1) = -3.$$

$$D_x = \det \begin{bmatrix} 7 & 1 \\ 2 & -1 \end{bmatrix} = 7(-1) - 1(2) = -9.$$

$$D_y = \det \begin{bmatrix} 2 & 7 \\ 1 & 2 \end{bmatrix} = 2(2) - 7(1) = -3.$$

Therefore

$$x = \frac{D_x}{D} = \frac{-9}{-3} = 3, \qquad y = \frac{D_y}{D} = \frac{-3}{-3} = 1.$$

21 Matrices as Transformations

21.1 A matrix moves vectors

A 2×2 matrix transforms vectors in the plane.

For example,

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$$

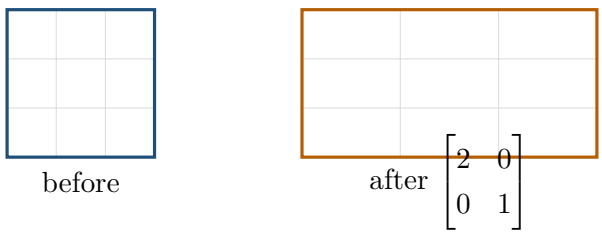
turns

$$\begin{bmatrix} x \\ y \end{bmatrix}$$

into

$$\begin{bmatrix} 2x \\ y \end{bmatrix}.$$

It stretches the x direction by a factor of 2 and leaves the y direction unchanged.



21.2 Rotation

A rotation by angle θ is represented by

$$R_\theta = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}.$$

For a rotation by $90°$,

$$\cos 90° = 0, \qquad \sin 90° = 1,$$

so

$$R_{90°} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$$

Then

$$R_{90^\circ}\begin{bmatrix}1\\0\end{bmatrix}=\begin{bmatrix}0\\1\end{bmatrix}.$$

The vector pointing right is rotated upward.

21.3 Why this matters in computer graphics

Computer graphics uses matrices for:

- scaling objects,
- rotating objects,
- projecting 3D scenes to 2D images,
- moving coordinate systems,
- composing transformations efficiently.

When transformations are represented by matrices, combining transformations becomes matrix multiplication.

22 Matrices as Data Structures in Computer Science

22.1 Images

A grayscale image can be represented as a matrix of pixel intensities:

$$\begin{bmatrix}0 & 20 & 40\\50 & 80 & 120\\200 & 220 & 255\end{bmatrix}.$$

Each entry is a brightness value. Image processing often means applying operations to such matrices.

22.2 Graphs

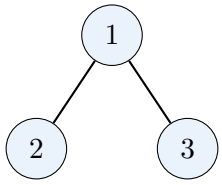
A graph can be represented by an adjacency matrix. Suppose we have three nodes. The matrix

$$A=\begin{bmatrix}0 & 1 & 1\\1 & 0 & 0\\1 & 0 & 0\end{bmatrix}$$

means:

- node 1 is connected to nodes 2 and 3,

- node 2 is connected to node 1,
- node 3 is connected to node 1.



22.3 Machine learning

Datasets are often matrices. Rows can represent observations and columns can represent features:

$$X = \begin{bmatrix} \text{user 1 feature 1} & \text{user 1 feature 2} & \cdots \\ \text{user 2 feature 1} & \text{user 2 feature 2} & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}.$$

Linear models use matrix expressions such as

$$X\mathbf{w} = \mathbf{y}.$$

This is one reason linear algebra is a foundation of machine learning.

23 Numerical Thinking: Exact Math vs Computer Math

23.1 Computers use approximations

In pure algebra, we can write exact fractions such as

$$\frac{1}{3}.$$

A computer often stores this approximately:

$$0.333333333333\ldots$$

This means numerical algorithms must care about rounding errors.

23.2 Why not always compute the inverse?

It is tempting to solve

$$A\mathbf{x} = \mathbf{b}$$

by writing

$$\mathbf{x} = A^{-1}\mathbf{b}.$$

But in numerical computing, explicitly computing A^{-1} can be slower and less stable than solving the system directly.

In practice, libraries usually use methods based on elimination, factorization, or iterative algorithms.

Big idea

For theory, inverses are important. For computation, solving systems directly is usually preferred.

24 Algorithmic View of Gaussian Elimination

24.1 Pseudocode idea

Gaussian elimination can be described algorithmically:

Gaussian elimination idea

For each pivot column:

1. Choose a pivot row.
2. If needed, swap rows.
3. Use the pivot row to eliminate entries below the pivot.

After the matrix is triangular, solve by back substitution.

24.2 Why pivoting matters

If a pivot is zero, we cannot divide by it. We may need to swap rows.

Example:

$$\left[\begin{array}{cc|c} 0 & 1 & 2 \\ 3 & 4 & 5 \end{array}\right].$$

The first entry is zero, so swap rows:

$$\left[\begin{array}{cc|c} 0 & 1 & 2 \\ 3 & 4 & 5 \end{array}\right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{cc|c} 3 & 4 & 5 \\ 0 & 1 & 2 \end{array}\right].$$

Now the pivot is 3, not zero.

24.3 Partial pivoting

In numerical computing, we often choose the largest available pivot in absolute value. This helps reduce numerical error.

This technique is called partial pivoting.

25 Common Beginner Mistakes

Common mistake

Mistake 1: Adding rows incorrectly.
If $R_2 \leftarrow R_2 - 2R_1$, then every entry of row 2 changes, including the right-hand side.

Common mistake

Mistake 2: Forgetting the right-hand side.
In an augmented matrix, row operations must be applied to both the coefficient part and the augmented column.

Common mistake

Mistake 3: Treating matrix multiplication like ordinary multiplication.
Usually $AB \neq BA$.

Common mistake

Mistake 4: Thinking determinant zero means no solution.
A zero determinant means not exactly one solution. The system may have no solution or infinitely many solutions.

Common mistake

Mistake 5: Dividing by zero.
If a pivot is zero, swap rows if possible. Do not divide by zero.

26 Mini Reference Sheet

26.1 Matrix sizes

$$(m \times n)(n \times p) = (m \times p).$$

26.2 Matrix-vector multiplication

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} ax + by \\ cx + dy \end{bmatrix}.$$

26.3 2×2 determinant

$$\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc.$$

26.4 2×2 inverse

If $ad - bc \neq 0$, then

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

26.5 Row operations

1. Swap rows.
2. Multiply a row by a nonzero scalar.
3. Add a multiple of one row to another row.

26.6 System outcomes

Situation after elimination	Meaning	Outcome
Pivot in every variable column	all variables determined	one solution
Row $0 = 0$ and free variables	some variables free	infinitely many solutions
Row $0 = c$, $c \neq 0$	contradiction	no solution

27 Practice Problems

27.1 Basic matrices and vectors

Practice

1. State the size of $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$.
2. For $A = \begin{bmatrix} 2 & -1 & 5 \\ 0 & 3 & 7 \end{bmatrix}$, find a_{12}, a_{23}, and a_{21}.
3. Compute $\begin{bmatrix} 2 \\ -3 \\ 5 \end{bmatrix} + \begin{bmatrix} 1 \\ 4 \\ -2 \end{bmatrix}$.
4. Compute $-2 \begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix}$.

27.2 Matrix operations

Practice

1. Compute $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} + \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$.
2. Compute $3\begin{bmatrix} 2 & -1 \\ 0 & 4 \end{bmatrix}$.
3. Compute $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}\begin{bmatrix} 5 \\ 6 \end{bmatrix}$.
4. Compute $\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} 3 & 4 \\ 5 & 6 \end{bmatrix}$.

27.3 Systems

Practice

Solve by elimination:

1. $\begin{cases} x + y = 8, \\ x - y = 2. \end{cases}$
2. $\begin{cases} 2x + y = 5, \\ x + 3y = 10. \end{cases}$
3. $\begin{cases} x + y + z = 6, \\ x - y + z = 2, \\ 2x + y - z = 3. \end{cases}$

27.4 Determinants

Practice

Compute the determinants:

1. $\det \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$
2. $\det \begin{bmatrix} 2 & 5 \\ 1 & 3 \end{bmatrix}$
3. $\det \begin{bmatrix} 1 & 2 & 0 \\ 0 & 3 & 4 \\ 0 & 0 & 5 \end{bmatrix}$

27.5 Concept questions

Practice

Answer in one sentence:

1. What does it mean if a 2×2 determinant is zero?
2. Why do we use an augmented matrix?
3. What is a pivot?
4. Why is $AB = BA$ not generally true?

28 Solutions to Selected Practice Problems

28.1 Basic matrices and vectors

1. The matrix $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ has size 2×3.
2. $a_{12} = -1$, $a_{23} = 7$, $a_{21} = 0$.
3. $\begin{bmatrix} 2 \\ -3 \\ 5 \end{bmatrix} + \begin{bmatrix} 1 \\ 4 \\ -2 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \\ 3 \end{bmatrix}$.
4. $-2 \begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} -6 \\ 0 \\ 2 \end{bmatrix}$.

28.2 Matrix operations

1. $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} + \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} = \begin{bmatrix} 6 & 8 \\ 10 & 12 \end{bmatrix}$.
2. $3 \begin{bmatrix} 2 & -1 \\ 0 & 4 \end{bmatrix} = \begin{bmatrix} 6 & -3 \\ 0 & 12 \end{bmatrix}$.
3. $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 1 \cdot 5 + 2 \cdot 6 \\ 3 \cdot 5 + 4 \cdot 6 \end{bmatrix} = \begin{bmatrix} 17 \\ 39 \end{bmatrix}$.
4. $\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 3 & 4 \\ 5 & 6 \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ 11 & 14 \end{bmatrix}$.

28.3 Systems

1. $$\begin{cases} x + y = 8, \\ x - y = 2 \end{cases}$$

   Add the equations:
   $$2x = 10 \Rightarrow x = 5.$$
   Then $5 + y = 8$, so $y = 3$. Solution: $(5, 3)$.

2. $$\begin{cases} 2x + y = 5, \\ x + 3y = 10. \end{cases}$$

   From the first equation, $y = 5 - 2x$. Substitute:
   $$x + 3(5 - 2x) = 10.$$

$$x + 15 - 6x = 10.$$

$$-5x = -5 \Rightarrow x = 1.$$

Then $y = 5 - 2 = 3$. Solution: $(1, 3)$.

3. The augmented matrix is

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & -1 & 1 & 2 \\ 2 & 1 & -1 & 3 \end{array}\right].$$

Subtracting the second equation from the first gives

$$2y = 4,$$

so

$$y = 2.$$

Then the first equation gives

$$x + z = 4.$$

The third equation becomes

$$2x + 2 - z = 3,$$

so

$$2x - z = 1.$$

Solving

$$x + z = 4, \qquad 2x - z = 1$$

gives

$$3x = 5,$$

hence

$$x = \frac{5}{3}, \qquad z = \frac{7}{3}.$$

The solution is

$$(x, y, z) = \left(\frac{5}{3}, 2, \frac{7}{3}\right).$$

28.4 Determinants

1. $\det \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = 1 \cdot 4 - 2 \cdot 3 = -2.$
2. $\det \begin{bmatrix} 2 & 5 \\ 1 & 3 \end{bmatrix} = 2 \cdot 3 - 5 \cdot 1 = 1.$
3. The matrix is triangular, so the determinant is $1 \cdot 3 \cdot 5 = 15$.

29 Final Learning Path

If a student is completely lost, use this order:

1. Learn to read matrix entries: row first, column second.
2. Practice matrix addition and scalar multiplication.
3. Practice matrix-vector multiplication until it becomes automatic.
4. Translate systems into augmented matrices.
5. Learn row operations carefully.
6. Solve 2×2 systems by elimination.
7. Solve 3×3 systems only after 2×2 systems are comfortable.
8. Learn determinants for 2×2 matrices.
9. Connect determinant zero with non-unique behavior.
10. Learn inverses only after determinants and elimination are clear.

Big idea

The main skill is not memorizing formulas. The main skill is translating a problem into rows, columns, pivots, and operations that preserve the solution.